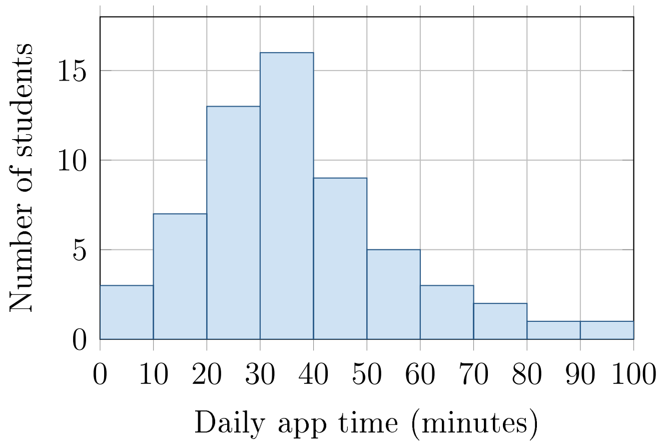
Check the Model

Topic 1.10 · TODAY'S PROBLEM

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes.

Ben: "About 16% should be above 52 minutes." **Ava:** "For 70 minutes, $z = 2.5$, so less than 1% should be above 70."



FIRST TAKE (5 min, on your sheet)

Does the normal model fit? Point to one histogram feature.

DOK 1 **(a)** State μ and σ for the proposed model. What two shape features should approximately normal data have?

DOK 2 **(b)** Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

DOK 3 **(c)** Are the calculations warranted as conclusions about these students? Defend your decision with the histogram's shape and tail count. Explain how the relationship between a model and its data supports your decision, what a reader would conclude about long app times, and what must be checked before using normal methods.

Video + follow-along: `u1_lesson10_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.